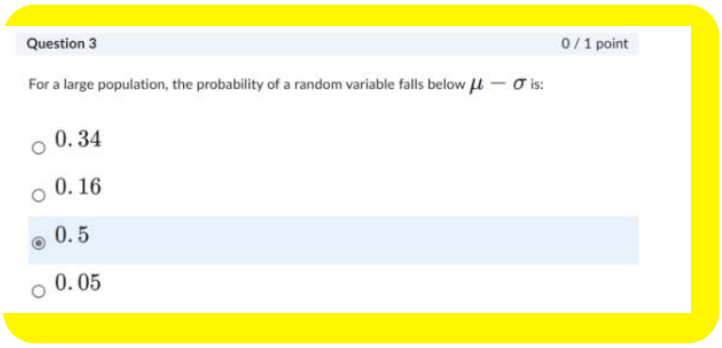
ChatGPT

Probability Below μ − σ

2/11/2026, 10:48:18 AM

Source: https://chatgpt.com/c/698ca492-7688-832c-9a05-4e107438e308



Okay Vasu, let's understand this **clearly and simply**.

The question:

> What is the probability that a random variable falls below **μ − σ**?

This means:

- μ = mean
- σ = standard deviation
- μ − σ = one standard deviation below the mean

Step 1: Convert to Z-score

In standard normal distribution:

$$Z = \frac{X - \mu}{\sigma}$$

If $X = \mu - \sigma$:

$$Z = \frac{(\mu - \sigma) - \mu}{\sigma} = \frac{-\sigma}{\sigma} = -1$$

So the question becomes:

> What is $P(Z < -1)$?

Step 2: Use standard normal values

From the Z-table:

$$P(Z < -1) = 0.1587 \approx 0.16$$

Final Answer:

✅ **0.16**

Why 0.5 is wrong?

0.5 is the probability below the **mean (μ)**.

But this question asks for **μ − σ**, which is one standard deviation below the mean — so the probability must be **less than 0.5**.

Quick Memory Trick (Empirical Rule 68–95–99.7 Rule)

- 68% is between μ − σ and μ + σ
- That leaves 32% outside
- Half of 32% is 16% on the left side

So:

$$P(X < \mu - \sigma) \approx 0.16$$

If you want, I can also explain with a simple graph to make it 100% clear 😊